

Assignment

Goal

Build a Go-based backend system that processes uploaded EDI-like files asynchronously and deploys locally using Docker Compose and Kubernetes.

Eg. Input EDI file:

```
CLAIM*CLM001*MEM123*2500
CLAIM*CLM002*MEM456*1800
CLAIM*CLM003*MEM123*300
```

Expected Json Output

```
{
  "claims": [
    {
      "claim_id": "CLM001",
      "member_id": "MEM123",
      "amount": 2500
    },
    {
      ...
    }...
  ],
  "summary": {
    "total_claims": 3,
    "total_amount": 4600
  }
}
```

Tech Stack

Go (Gin/Echo/net-http), MongoDB, Redis, Docker, Docker Compose, Kubernetes (Minikube/Kind/Docker Desktop Kubernetes).

Functional Requirements

1. POST /jobs: Upload file and create job.
2. GET /jobs/{job_id}: Fetch job status.
3. GET /jobs/{job_id}/result: Fetch parsed result.

Async Processing Flow

Upload → Store → Create job → Push to Redis queue → Worker processes → Store result.

Services Required

API Service, Worker Service, Redis, Database.

Docker Requirement

Provide docker-compose.yml to run all services locally.

Kubernetes Requirement

Provide manifests for deployments and services. Ensure the system runs locally. API should be accessible locally using port-forward

Engineering Requirements

Health endpoint, structured logging, graceful shutdown, env config, error handling, README(with complete documentation and deployment details).

Bonus

Retries, job failure handling, tests, metrics, Swagger.

Timeline

Complete within 24 hours.